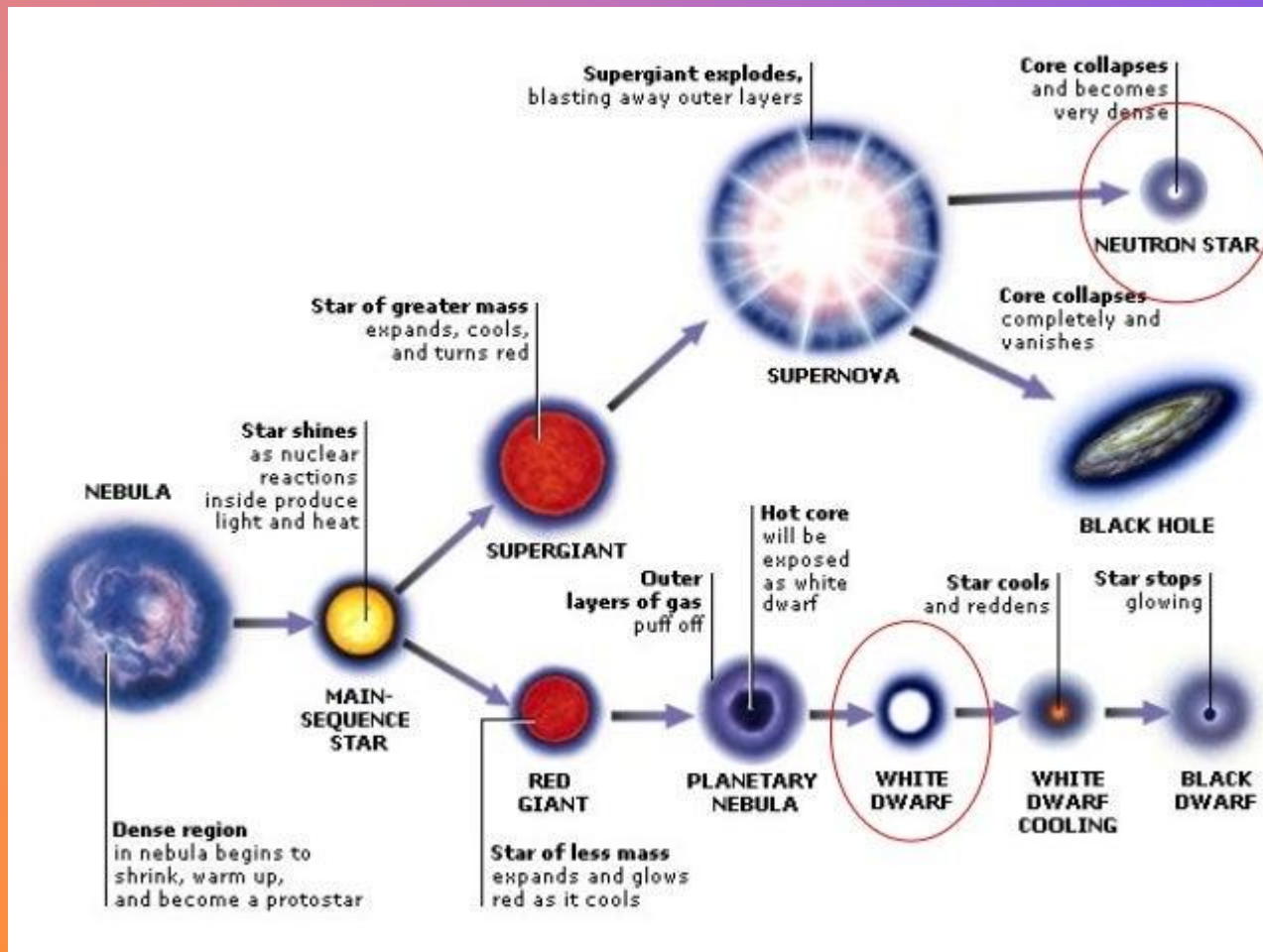


NEUTRON STARS AND EQUATIONS OF STATE



WHAT IS A NEUTRON STAR?



- Compact remnant after a supernova
- Mass around $1-2 M_{\odot}$
- Radius around 10-14 km
- Extremely dense, strong gravity
- Equation of state: $p=p(\epsilon)$
- Determines neutron star structure
- Soft EOS $\rightarrow$ lower pressure
- Stiff EOS $\rightarrow$ higher pressure
 - Larger radii
 - Higher maximum masses

TOV EQUATIONS

- Relativistic hydrostatic equilibrium
- Pressure gradient, enclosed mass
- Pressure contributes to gravity in GR



A diagram showing a central point with three concentric circles around it. The innermost circle is solid black, the middle one is dashed black, and the outermost one is solid black. This represents the layers of a star.

$$\frac{dP}{dr} = \frac{-G(\varepsilon + P/c^2)(m + 4\pi r^3 P/c^2)}{r(r - 2Gm/c^2)}$$

$$\frac{dm}{dr} = 4\pi r^2 \varepsilon$$

Numerical idea:

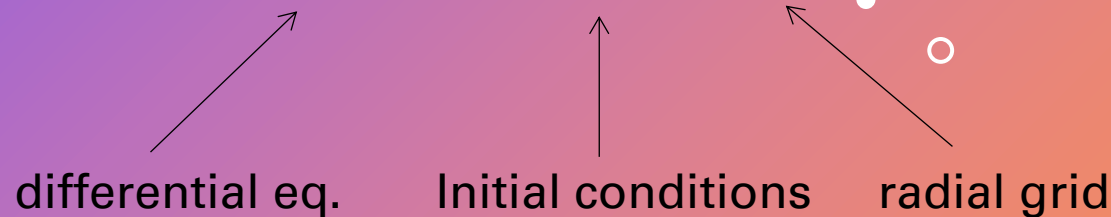
- Choose central density, compute central pressure
- Integrate outward, stop at stellar surface
- Obtain mass and radius
- Repeat for many central densities
- Show in code: line 64

EOS IN OUR CODE

- `self.en_dens = interp1d(...)` – line 39, creates and stores a continuous EOS interpolation function
- `eden = self.en_dens(P)` – line 70, uses this EOS function during the TOV integration
- During integration -> corresponding energy density to given pressure

Where does integration happen?

- `psol = odeint(self.tov_eq, [P, m], r, rtol=rtol)` – line 92



CONSTRUCTING THE MR DIAGRAM

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- for cd in cd_grid:
 try:
 R, M = tov.solve(cd, dr=dr) – line 153
- One central density -> one neutron star
- Repeating the calculation builds the MR curve
- **TASK 1:**
 - Create an array of central densities between cd_min and, cd_max using cd_points – line 145

CONSTRUCTING THE MR DIAGRAM



- for `cd` in `cd_grid`:
 try:
 `R, M = tov.solve(cd, dr=dr)` – line 153
- One central density -> one neutron star
- Repeating the calculation builds the MR curve
- **TASK 1:**
 - Create an array of central densities between `cd_min` and `cd_max` using `cd_points` – line 145
 - **Solution:** `cd_grid = np.linspace(cd_min, cd_max, cd_points)`
 - Each `cd` -> one NS configuration
 - More points -> smoother MR diagram, longer runtime

CONSTRUCTING THE MR DIAGRAM



- **TASK 2:** Find the index of the maximum mass – line 180
 - `i_max =...`

CONSTRUCTING THE MR DIAGRAM

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- **TASK 2:** Find the index of the maximum mass – line 180
 - `i_max = ...`

Solution:

- `i_max = np.argmax(masses_all)`
- Maximum mass separates stable and unstable solutions
- `np.argmax()` returns the index, not the value itself -> index used to split all arrays

TASK 3: split the MR diagram into stable and unstable branch – line 187

- `radii_stable = ...`
- `masses_stable = ...`
- `Pc_stable = ...`

- `radii_unstable = ...`
- `masses_unstable = ...`

- Unstable after maximum mass

TASK 3: split the MR diagram into stable and unstable branch – line 187

- `radii_stable = ...`
- `masses_stable = ...`
- `Pc_stable = ...`
- `radii_unstable = ...`
- `masses_unstable = ...`
- Unstable after maximum mass

SOLUTION:

- `radii_stable = radii_all[:i_max + 1]`
- `masses_stable = masses_all[:i_max + 1]`
- `Pc_stable = Pc_used[:i_max + 1]`
- `radii_unstable = radii_all[i_max + 1:]`
- `masses_unstable = masses_all[i_max + 1:]`
- `i_max + 1` includes the max mass point

- **TASK 4:** find: maximum mass, radius at max mass – line 197

- $M_{\text{max}} = \dots$

- $R_{\text{at_Mmax}} = \dots$



- **TASK 4:** find: maximum mass, radius at max mass – line 197

- `M_max = ...`

- `R_at_Mmax = ...`



SOLUTION:

- `M_max = masses_all[i_max]`
- `R_at_Mmax = radii_all[i_max]`
- `M_max` is the largest mass supported by the EOS
- Configurations after the max mass are unstable

TASK 5: find the radius of a neutron star with $M=1.4 M_{\odot}$ – line 206

- $R_{\text{of_M}} = \dots$
- $R_{14} = \dots$

TASK 5: find the radius of a neutron star with $M=1.4 M_{\odot}$ – line 206

- `R_of_M = ...`
- `R_14 = ...`

SOLUTION:

- `R_of_M = interp1d(`
- `masses_stable,`
- `radii_stable,`
- `bounds_error=False,`
- `fill_value=np.nan`
- `)`
- `R_14 = float(R_of_M(1.4))`
- `R_14` is one of the most important observable NS parameters
- `bounds_error=False` prevents the interpolation from crashing when the requested value is outside the known range
- `fill_value=np.nan` returns “not a number” for invalid interpolation requests

TASK 6: mark the max mass in the MR diagram using a red (or other color) point – line 278

- `plt.scatter(...)`

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TASK 6: mark the max mass in the MR diagram using a red (or other color) point – line 278

- `plt.scatter(...)`

SOLUTION:

- `plt.scatter(`
- `R_at_Mmax,`
- `M_max,`
- `color="red",`
- `s=80,`
- `label="maximum mass"`
- `)`

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FINAL MR DIAGRAM

Standard MR diagram:

- Stable branch
- Unstable branch
- Max mass

Colored MR diagram:

- Color given by central pressure
- `Pc_log = np.log10(Pc_stable)`
- Central pressure spans several orders of magnitude -> use a logarithmic scale to visualize the color gradient more clearly



INTERPOLATION EXPERIMENT

- Try to experiment with interpolation methods – line 25
- Which gives the smoothest curve?
- Which creates suspicious behaviour?
- Does smooth always mean physical?

INTERPOLATION EXPERIMENT

- Try to experiment with interpolation methods – line 25
- Which gives the smoothest curve?
- Which creates suspicious behaviour?
- Does smooth always mean physical?

Typical behaviour:

- Nearest -> steps
- Linear -> robust
- Quadratic -> smoother
- Cubic -> smooth, may oscillate

Numerical accuracy experiment:

- Experiment with the integration step size – line 21
- $dr = 20$
- What happens to the MR curve?
- Does runtime change?
- Unstable solution?

Numerical accuracy experiment:

- Experiment with the integration step size – line 21
- $dr = 20$
- What happens to the MR curve?
- Does runtime change?
- Unstable solution?

-> Numerical calculations always involve a tradeoff between accuracy and computational speed

Smaller dr

- More accurate
- Smoother solution
- Slower calculation

Larger dr

- Faster calculation
- Lower accuracy
- Possible numerical instability

CONCLUSION

Physics:

- NS structure depends on the EOS
- TOV describe relativistic stellar equilibrium
- Different EOS produces the MR diagram



Numerical methods:

- Interpolation affects the EOS shape
- Integration step size affects accuracy
- Numerical methods influence the final MR diagram